United Airlines Departure Delays: Written Report

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Team Contributions

- Binay Raut: Data preparation, exploration data analysis, statistical analysis, visualizations, preparation of report and PowerPoint slide deck.
- Nina Fossen: Partial contribution to the project, including project planning, preliminary data preparation, and exploratory analysis.

Introduction

United Airlines is committed to improving efficiency and customer satisfaction. Departure delays compromise efficiency and erode customer satisfaction over time. To decrease departure delays for United Airlines flights, it is crucial to first investigate what may be affecting the delays. A better understanding of factors associated with departure delays is an important step toward identifying opportunities for improvement. This report investigates departure delays for United Airlines flights departing from 3 New York area airports: John F. Kennedy (JFK), LaGuardia Airport (LGA), and Newark Liberty (EWR) by examining how weather and time of day/year may impact departure delays. Using flight and weather data from the nycflights13 package in R, we examine the relationship between departure delays and six factors: time of day, time of year, temperature, wind speed, precipitation, and visibility were explored. Our goal is to use exploratiory data analysis and permutation tests to identify patterns and determine whether observed differences in departure delays are statistically significant.

Data Methodology

Data Source

The analysis used the *nycflights13* dataset. This dataset contains information about flights departing from the three airports in New York airports. The source dataset has information about airlines, scheduled departure time, actual departure time, and delays. Additionally, this analysis also used hourly weather data for each airport using the *weather* dataset.

This analysis focused on flights during the year 2013. Additionally, only United Airlines (UA) flights were considered for analysis.

Data Preparation

Filtering and merging

Initial data preparation included filtering data to only include flight observations from United Airlines (UA). Flights records from united airlines for the year 2013 were then combined with hourly weather observation data. This allowed us to expand our dataset to include temperature, wind speed, precipitation and visibility to each of our remaining flights observations.

Missing Data

Summary of Missing Values in UA Flight Data

Variable	Count
total_flights	58665
missing_hour	0
missing_month	0
missing_dep_delay	686
missing_temp	311
missing_wind_speed	321
missing_precipitation	304
missing_visibility	304

TABLE 1: TOTAL OBSERVATIONS AND TOTAL MISSING OBSERVATIONS FOR KEY MEASURES

We were missing 686 departure delays from the dataset. Since we are analyzing departure delays, we removed any observations where departure delays were missing. After filtering observations based on airline and missing departure delays, we had 57,979 observations remaining in our dataset.

Although observations were missing other weather-related data, we decided to keep it from our analysis dataset. Missing weather values were handled separately during analysis.

Data Augmentation

We added two indicators related to departure delays using available data for departure delays. Firstly, we added a column that captured is a flight was late to depart. A flight departing more than 0 minutes after scheduled departure time would be considered late. Finally, we added a column that captured if a flight was very late to depart. A flight

departing more than 30 minutes after its original scheduled departure time would be considered very late.

Analysis

We explored patterns in departure delays based on various factors like time of day (hour), time of year (month), as well as weather related factors like wind speed, temperature, precipitation and visibility. We then conducted statistical testing to further examine departure delays.

For statistical testing, we selected outcome measures based on the research question being investigated. We compared mean departure delays between morning and evening flights to examine differences in delay duration. For the remaining five factors, we compared the proportions of late and very late departures to examine the frequency and severity of delays under different conditions. Two-sided permutation tests were used to evaluate whether the observed differences were statistically significant.

Results

For this analysis, we examined six factors that could potentially influence or explain departure delays for United Airlines flights departing New York area airports.

- Time of day
- Time of year
- Temperature
- Wind Speed
- Precipitation
- Visibility

Time of day

Flights scheduled during the early morning generally experience shorter average delays, while flights scheduled later in the day tend to experience longer delays. For example, flights scheduled at 6 AM had an average departure delay of 2.72 minutes, compared with 22.60 minutes for flights scheduled at 6 PM.

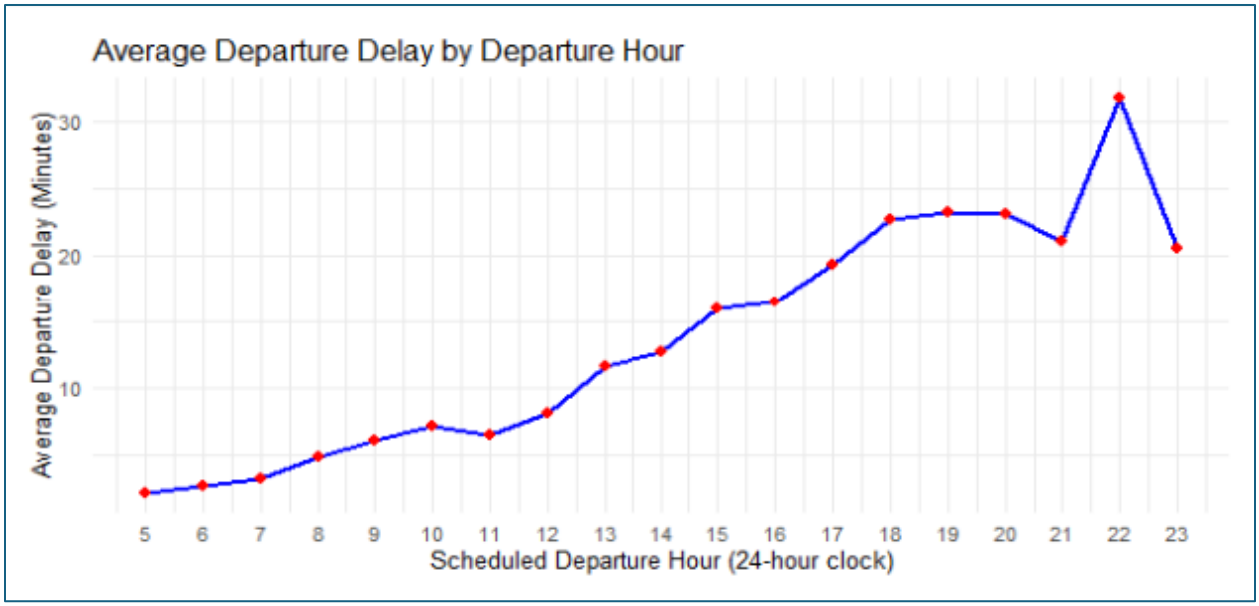


Figure 1: Line chart showing average departure delays by hour

A similar pattern was observed when examining the proportion of late and very late flights. At 6 AM, 30.42% of flights departed late and 4.44% departed more than 30 minutes late. By 6 PM, these percentages increased to 64.13% and 23.89%, respectively.

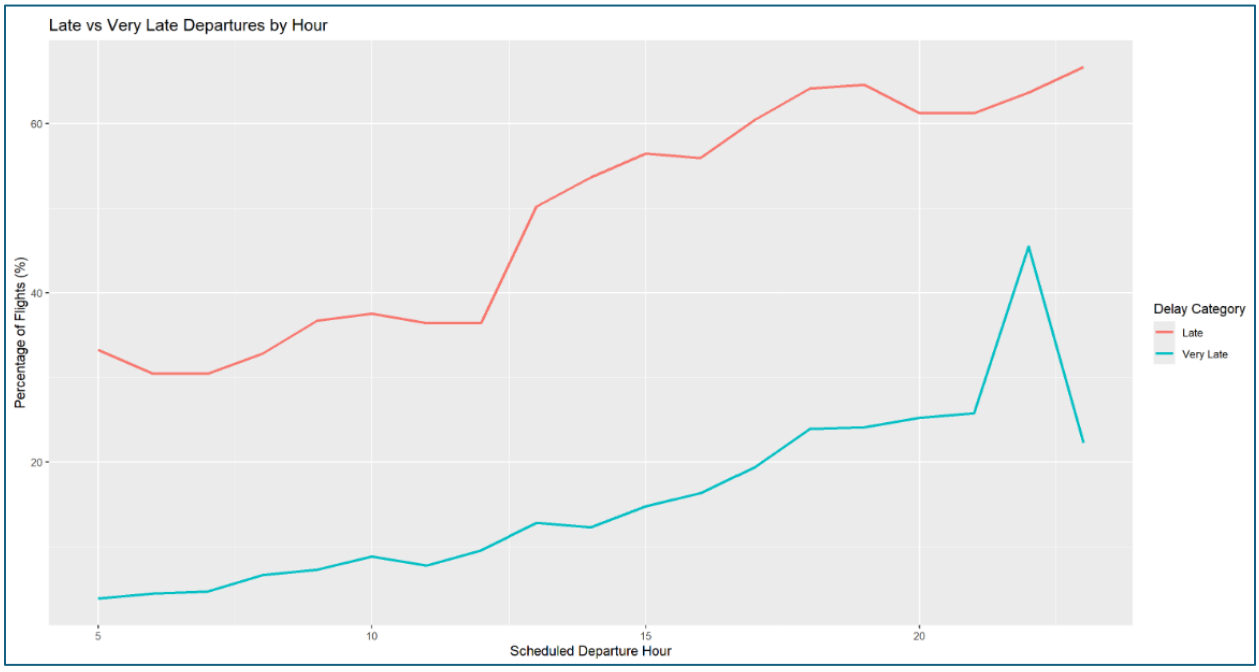


Figure 2: Line chart comparing Late vs Very Late flights by departure hour

Statistical analysis found evidence of differences in average departure delays between morning and evening flights. The analysis indicates that departure delays for flights

departing late afternoon and evenings are higher than for flights departing early in the morning.

Time of year

Number of flights is approximately evenly distributed between all 12 months. Departure delays varied throughout the year. June and July experienced the longest average departure delays, at approximately 20 minutes.

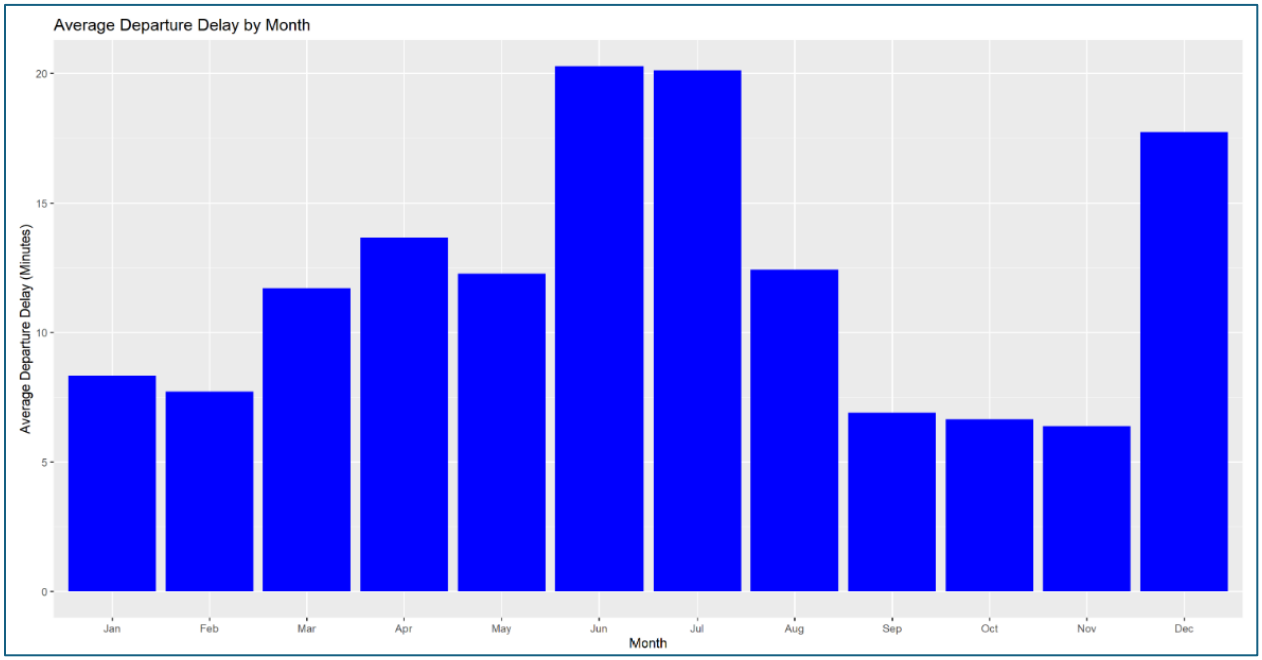


Figure 3: Bar chart showing average departure delay by month

Additionally, June and July also had one of the highest percentages of late and very late departures, with over 56% of flights departing late and approximately 20% of flights departing more than 30 minutes late. The month of December also had a high percentage of delays with an average delay of 17.72 minutes. December had the highest percentage of flights late with 64.68% of flights late.

September, October, and November experienced shorter average delays of around 6 to 7 minutes and lower rates of late and very late delays. The analysis shows that departure delays and proportion of late and very late delays are higher during early summer months and December. We compared summer months (June, July and August) with winter months (December, January and February). Statistical testing found strong evidence that departure delays rates during summer were higher than during winter although December experienced especially high rate of delays. We found that late departures during summer flights were 2.48 percentage points higher and very late departures (over 30 minutes late) around 5% higher during summer months.

Temperature

To analyze departure delays and temperature, we grouped temperature into three categories: Cold (temperature less than 40 degrees), Moderate (temperature between 40 and 80) and Hot (temperature over 80). Flights departing under hot conditions experienced the longest average delays, approximately 20.27 minutes, compared with 11.23 minutes under moderate conditions and 9.94 minutes under cold conditions.

Hot conditions also had the highest proportions of late and very late departures, at 57.62% and 19.25%, respectively. In contrast, moderate temperatures had the lowest proportion of late departures, at approximately 44.69%. These findings suggest an association between higher temperatures and increased departure delays.

Departure Delays by Temperature Category

Temperature Category	Total Flights	Mean Delay (min)	Median Delay (min)	Late (%)	Very Late (%)
Cold	13033	9.94	0	47.43	11.09
Hot	7371	20.27	3	57.62	19.25
Moderate	37275	11.23	0	44.69	12.61

Table 2: Flight delays by Temperature Category

Statistical analysis found strong evidence of difference between hot and moderate temperature. Flights departing during hot conditions had 12.93 percentage points higher rate of late departure. Additionally, flights departing during hot conditions had 6.64 percentage points higher rate of very late departures compared to moderate conditions.

Wind Speed

The wind speed analysis showed that most United Airlines flights departed under wind conditions below 20 mph. When wind speeds were grouped into category: low (less than 10mph), moderate (between 10 to 20), and high (more than 20). Average departure delays were 11.26 minutes during low winds, 13.01 minutes during moderate winds, and 12.29 minutes during high winds.

Departure Delays by Wind Category

Wind Category	Total Flights	Mean Delay (min)	Median Delay (min)	Late (%)	Very Late (%)
low	29080	11.26	0	45.04	12.37
moderate	25961	13.01	0	48.93	13.81
high	2628	12.29	0	48.86	14.65

Table 3: Departure Delays by Wind Category

The percentage of late departures was slightly higher under moderate and high wind conditions compared with low wind conditions. Additionally, the proportion of flights departing more than 30 minutes late increased from 12.37% under low wind conditions to 14.65% under high wind conditions.

For statistical testing, wind speeds were grouped into two categories: low (below 10 mph) and higher (10 mph or greater). The number of flights was approximately evenly distributed between the two groups.

Statistical testing found evidence of differences in late and very late departure rates between the groups. However, the differences observed were relatively small. Flights experiencing higher wind speeds had a 3.89% higher rate of late departures and a 1.52% higher rate of very late departures compared with flights experiencing lower wind speeds.

Overall, higher wind speeds were associated with slightly more frequent departure delays, although the differences were modest.

Precipitation

Departure delays were compared between dry conditions, defined as hours with no recorded precipitation, and wet conditions, defined as hours with measurable precipitation. The precipitation analysis showed that most United Airlines flights departed during dry conditions. Of the 57,686 flights with available precipitation data, approximately 93% occurred when no precipitation was recorded, while 7% occurred under wet conditions. Flights scheduled during wet conditions experienced an average departure delay of 24.63 minutes, compared with 11.15 minutes during dry conditions.

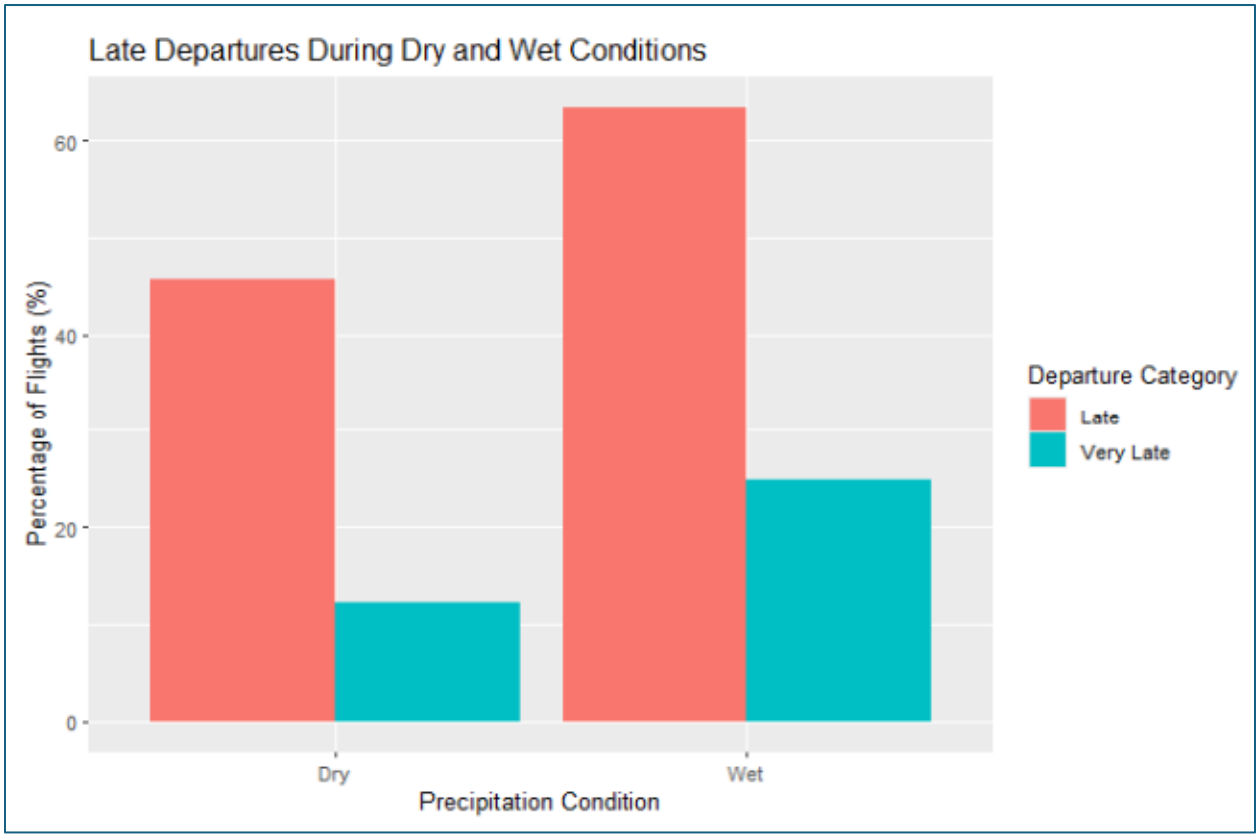


Figure 4: Late departures during dry and wet conditions

The proportion of late departures increased from 45.72% under dry conditions to 63.42% under wet conditions. Similarly, the proportion of flights departing more than 30 minutes late increased from 12.24% during dry conditions to 24.80% during wet conditions.

Statistical analysis found strong evidence of difference in late and very late departure delays between dry and wet conditions. Flights experiencing precipitation had a 17.7 percentage points higher rate of later departures. Finally, flights experiencing precipitation had 12.55 percentage points higher rate of very late departures.

Visibility

Of 57,686 flights with available visibility observations, approximately 83% occurred under normal recorded visibility (10 miles), while the remaining 17% experienced reduced visibility (below 10 miles). Flights operating under reduced visibility experienced longer average departure delays of approximately 17.03 minutes, compared with 11.08 minutes under normal recorded visibility. Additionally, the proportion of late departures was also higher under reduced visibility, at 51.31%, compared with 46.08% under normal visibility. Similarly, very late departures occurred in 17.47% of flights under reduced visibility, compared with 12.23% under normal visibility.

Departure Delays by Visibility Condition

Visibility Condition	Total Flights	Mean Delay (min)	Median Delay (min)	Late (%)	Very Late (%)
Normal	47888	11.08	0	46.08	12.23
Reduced	9798	17.03	1	51.31	17.47

Figure 5: Departure delays by visibility condition

Statistical analysis found strong evidence of difference between flight delays under normal and reduced visibility. Flights experiencing reduced visibility had 5.23 percentage points higher rate or late and 5.24 percentage points higher rate of very late departures. Reduced visibility was associated with frequent departure delays.

Discussion

The goal of this analysis was to study departure delays for the flights departing New York's 3 airports in 2013 to improve customer satisfaction. We examined relationship between delays and 6 factors: time of day, time of year (season), temperature, wind speed, precipitation, and visibility.

Time of day, precipitation and temperature had the most notable difference in departure delays. Flights departing in the Evening hours experienced an average delay of around 18 minutes longer than flights departing in the morning hours. Flights departing during precipitation or wet conditions had 17.7% higher rate of delayed departures compared to flights departing during dry conditions. Finally, flights departing during hot conditions had 12.93% higher rate of delays compared to flights leaving during moderate temperature.

Potential improvement in customer satisfaction can be made by accounting for known factors that may cause flight delays. For example, more flights can be scheduled during morning hours. Additionally, it may be worthwhile investigating operations during wet weather and high temperatures.

Challenges

The results of statistical tests and analysis should not be interpreted as evidence of cause-effect relationships. Our analysis examined each factor separately without accounting for potential relationships among the factors. For example, high temperatures are often associated with summer months, while precipitation and reduced visibility may occur together. Consequently, the observed differences may be influenced by multiple overlapping conditions.

Additionally, the analysis also focused on limited datasets involving only one city and one year of observations. The findings provide insights into departure delays within this specific dataset but may not generalize to other airports, airlines, or years.

Improvements and Extensions

The analysis could be expanded to include multiple years and data from other airports to determine if the observations from other airports and other years follow the same pattern. Additionally, other factors like air traffic, congestion, and data from destination could be used to help explain departure delays. Additional statistical analysis could be performed to better understand the association of factors with departure delays.

Broader Impact

A better understanding of delays can support a more informed data-driven decision. This analysis attempts to examine conditions that could indicate when departure delays are frequent and severe. The information obtained could be used to expand research further. By using the findings to guide further research and operational planning, United Airlines can identify opportunities to improve scheduling, as well as resource allocation. Ultimately this would help enhance passenger satisfaction.